

Stage 1 Tutorial Activity

- Why Software Engineering Still Matters

Stage 1 | Introducing Software Technology Case Study with Python and Guided AI use

Learning goals

- Explain why software engineering is broader than coding.
- Identify stakeholders in a simple software problem.
- Recognise missing requirements.
- Critically evaluate AI-generated feature suggestions.
- Explain why AI output should not automatically be treated as correct.

Activity 1 - Think-Pair-Share (10 minutes)

If ChatGPT or Copilot can produce a 100-line Python application very quickly, what knowledge does a software engineer still need?

1. AI can write the code quickly, but it cannot fully understand the problem. The software engineer still needs to gather the requirements, the needs of the clients, and decide whether the application is actually suitable for what it is intended for.
2. Software Engineers need to check if the code is correct, precise, or as described in how it was intended to be. As AI can make mistakes and only make the program based on what it thinks, and not based on what the engineer needs
3. Software Engineer's job is also to maintain a program, an AI cannot maintain the whole system bwwwwwy itself

Activity 2 - Is This Software Engineering? (10 minutes)

Scenario A: A student writes a 50-line Python calculator.

Scenario B: A team develops a payroll system used by 5,000 employees.

Scenario C: An AI assistant generates a simple appointment application from one prompt.

Scenario	Programming?	Software engineering?	Why?
A	Yes	No	It is programming because it involves writing code. But it is not considered software engineering. This is because it is only a small amount of code, and it does not include any maintenance, testing, or teamwork.

B	Yes	Yes	It is programming because making a payroll system will include writing code. It is also software engineering because this scenario involves having to plan, test, maintain the system in the long run, and it involves a real-world scenario
C	Yes	No	It is programming as it does involve the need to write a code. But it is not software engineering because generating a prompt in AI does not allow people to use planning, testing the code, maintenance in the long run. It also needs a software that you can use multiple times of any scenario, not just a onetime use

Activity 3 - SmartCare Problem Analysis (20 minutes)

Client statement: SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The clinic wants new software to improve these processes.

Task 1 - Identify stakeholders

Stakeholder	What do they need?
Patient	A simple way for them to book or cancel their next appointment
Doctors	They would need access to patient information, as well as access to medical supplements and equipment. And also, a way to update appointment and add notes
Receptionists	A fast system to quickly register patients, check patients, check availability of doctors, and updating patient details
Manager	A bigger system that allows them to track everything that is in the clinic. Such as having access to patients, doctors, medical supplies or equipment, records and reports, and controlled staff access

Task 2 - Identify current problems

1. Patient information or records might be lost, stolen, or misplaced when they are placed on a paper sheet. The paper can also be crumbled from when stored, making the information harder to read or possibly damaged. It also has limited backup for when an extra is needed

2. Staff may entry wrong data information. Such as writing down wrong patient details, and using outdated information

3. Appointments might be doubled or missed because they can lose track of the papers due to how many there are. Multiple staff members might also have different records of availabilities and such without seeing th most updated record

4. Patient information might not be secure because paper can be easily flown away, and spreadsheets can easily be accessed, copied or shared by other people.

Task 3 - Ask client questions

1. What information would you need about patients? Would you need their address, name, surname, phone number, etc.?

2. Who will use the system? Would all staff members have access? If so, what would they have access to or be able to view and edit?

3. Does the clinic need automatic reminders about upcoming appointments such as email or SMS?

4. What types of appointments does the clinic offer, and how long should each appointment be

5. What rules should be made to prevent double booking. Such as, should one doctor only allowed to have one patient per time slot?

Activity 4 - Critique an AI Response (15 minutes)

An AI assistant suggests: appointment management; facial-recognition login; AI diagnosis recommendations; patient search; online payment; practitioner schedule view; insurance processing; automatic treatment-plan generation.

Suggestion	Client evidence?	In scope?	Decision
Appointment management	The clinic needs to manage appointments more effectively	Yes	Include
Facial recognition login	The client did not ask for a biometric login	No	Exclude
AI diagnosis recommendations	The client only asked for patient appointment processes be quicker and more efficient	No	Exclude
Patient search	Staff needs to find and manage patient information	Yes	Include
Online payment	Payments during appointment were not mentioned, but can ask client	No	Ask if needed
Practitioner schedule view	Staff needs to see availability when making appointments	Yes	Include
Insurance processing	Not mentioned by client, but can ask as well	No	Ask if needed

Treatment-plan generation	Not needed by client since they asked for patient records system, and not for treatment decisions	No	No
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Exit question

Write one activity that a software engineer must perform and that cannot safely be delegated entirely to AI.

- A software engineer must talk with the stakeholders and figure out the problems and risks that they are facing, such as incorrect appointments or unauthorised access to patient information. They then need to decide the required action that they need to implement based on the given feedback. AI can suggest ideas, but a human engineer makes the decision and verifies them and remains accountable which are part of managing AI related risks.